Properties of the Fourier Transform and the Discrete-Time Fourier Transform

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Key Concepts

1) The Fourier transform (FT) and discrete-time Fourier transform (DTFT) have several key properties.

2) The FT and DTFT both convert convolution of time-domain signals to multiplication of Fourier representations.

    a) The FT of a convolution of signals $x(t)$ and $y(t)$ is the product of the FTs $X(\Omega)$ and $Y(\Omega)$.

    b) The DTFT of a convolution of signals $x[n]$ and $y[n]$ is the product of the DTFTs $X(e^{j\omega})$ and $Y(e^{j\omega})$.

c) Multiplication in frequency leads to the notion of filtering, or separating signals based on their frequency content.

3) The FT or DTFT of the product of time domain signals is the convolution of FTs or DTFTs, respectively.

a) The FT of the product $x(t)w(t)$ is the convolution $\left(\frac{1}{2\pi}\right) X(\Omega) * Y(\Omega)$.

b) The DTFT of the product $z[n] = x[n]w[n]$ is the 2π periodic convolution

$$Z(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\nu}) W(e^{j(\omega-\nu)}) d\nu$$

4) Shifting a signal in time introduces a linear phase shift in the FT or DTFT.

    a) If $y(t) = x(t - t_0)$, then the FT $Y(\Omega) = e^{-j\Omega t_0} X(\Omega)$.

    b) If $y[n] = x[n - n_0]$, then the DTFT $Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$.

5) The FT can be used to represent periodic signals. In this case the FT is a function of the Fourier series coefficients for the signal.

    a) Let $x(t)$ be periodic with period T, fundamental frequency $\Omega_0 = \frac{2\pi}{T}$ rads/s and Fourier series coefficients $X[k]$.

b) The FT representation for $x(t)$ is given by

$$X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} X[k]\delta(\Omega - k\Omega_0)$$

Video Screenshots

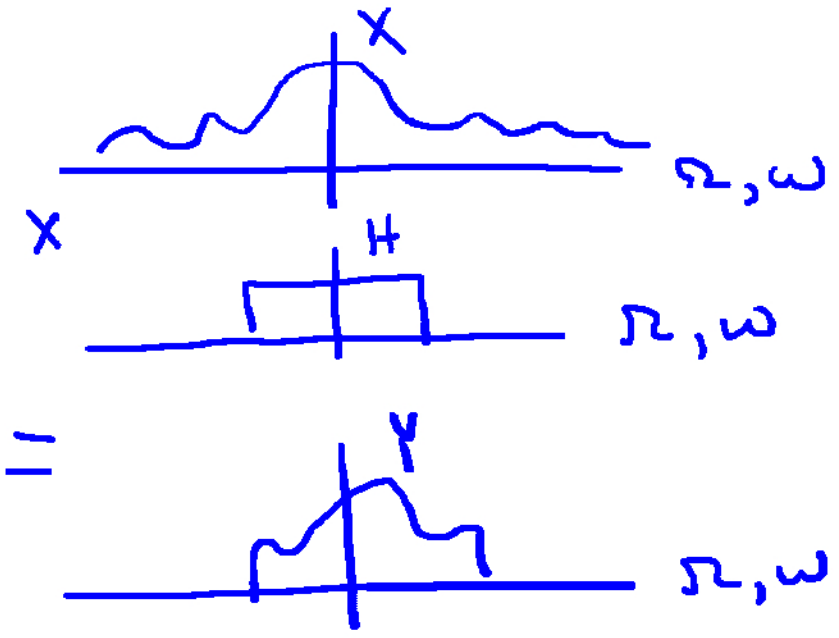
FT and DTFT Properties

A) Convolution – multiplication

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau \quad \xleftrightarrow{FT} \quad Y(\Omega) = X(\Omega)\, H(\Omega)$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] \quad \xleftrightarrow{DTFT} \quad Y(e^{j\omega}) = X(e^{j\omega})\, H(e^{j\omega})$$





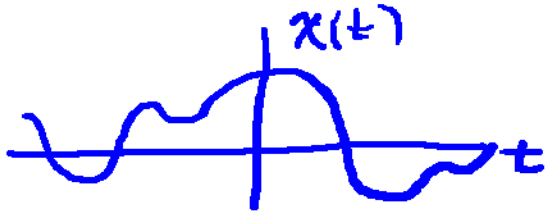
"Filtering"

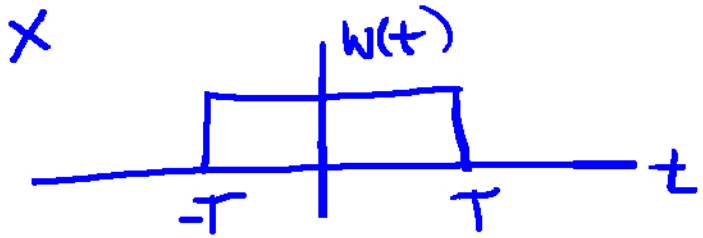
B) Multiplication – Convolution (windowing)

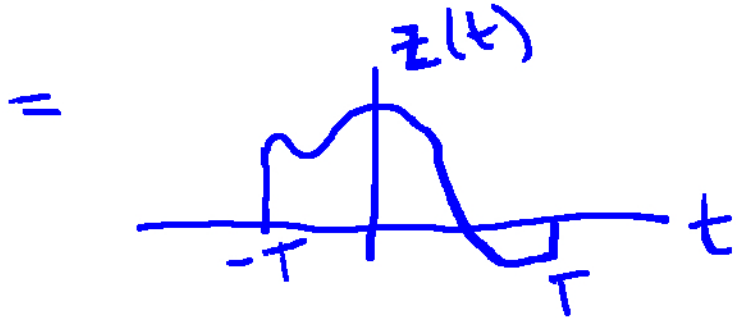
$$z(t) = x(t)\,w(t) \quad \overset{FT}{\longleftrightarrow} \quad Z(\Omega) = \frac{1}{2\pi} X(\Omega) * W(\Omega)$$

$$z[n] = x[n]\,w[n] \quad \overset{DTFT}{\longleftrightarrow} \quad Z(e^{j\omega}) = \frac{1}{2\pi} X(e^{j\omega}) \circledast W(e^{j\omega})$$

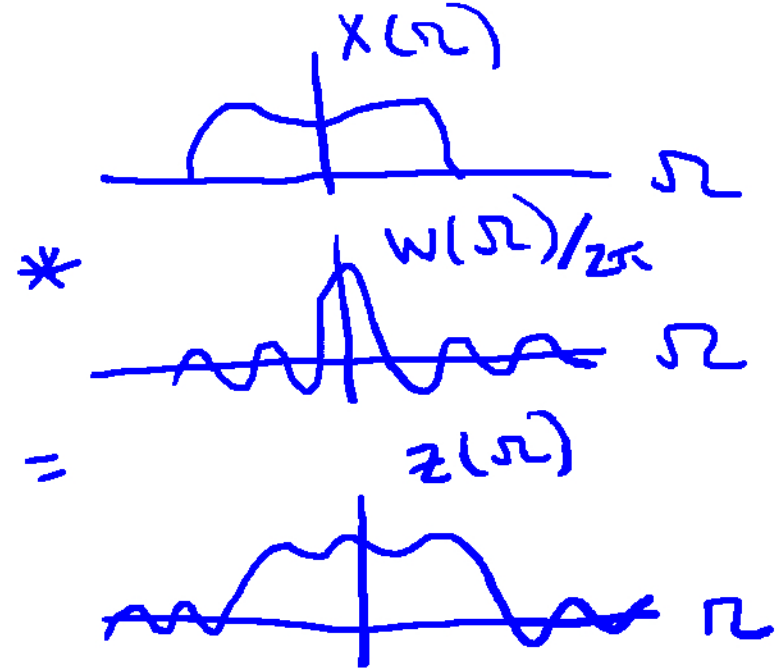
$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\nu})\, W(e^{j(\omega-\nu)})\, d\nu$$







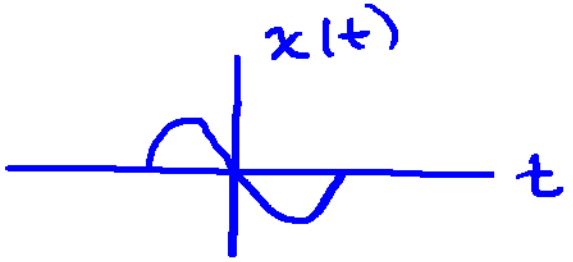




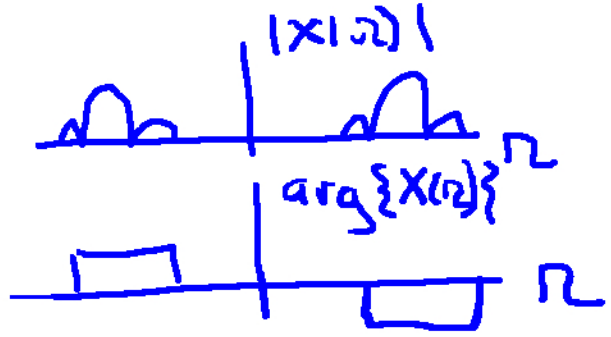
c) Time Shift

$$y(t) = x(t - t_0) \quad \overset{FT}{\longleftrightarrow} \quad Y(\Omega) = e^{-j\Omega t_0} X(\Omega)$$

$$y[n] = x[n - n_0] \quad \overset{DTFT}{\longleftrightarrow} \quad Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$$

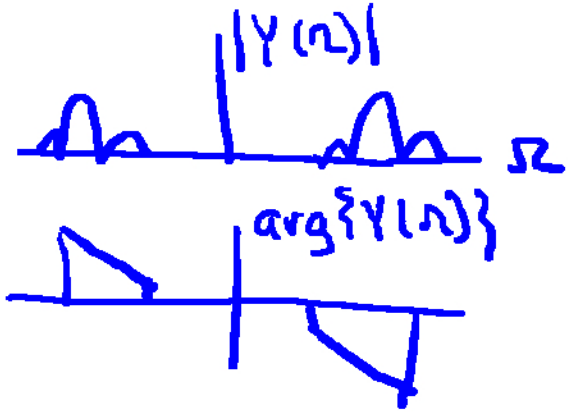


$\overset{FT}{\longleftrightarrow}$





$\overset{FT}{\longleftrightarrow}$

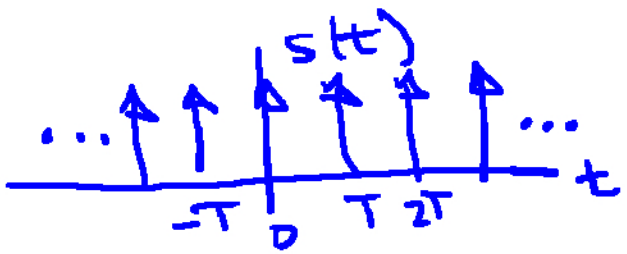


D) FT Representation for Periodic Signals

$x(t)$ has fund. period T: $x(t) \overset{FS;\Omega_0=\frac{2\pi}{T}}{\longleftrightarrow} X[k] = \frac{1}{T}\int_0^T x(t)e^{-jk\Omega_0 t}dt$

Then $x(t) \overset{FT}{\longleftrightarrow} X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} X[k]\,\delta(\Omega - k\Omega_0)$

Example:



$s(t) = \sum_{l=-\infty}^{\infty} \delta(t - lT)$

① Find FS

$S[k] = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)e^{-jk\frac{2\pi}{T}t}dt$

$= \frac{1}{T} \cdot 1 = \frac{1}{T}$

② $S(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T}\,\delta(\Omega - k\frac{2\pi}{T})$

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